

## CHAPTER 4

(15-25 pages..)

### SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

#### 4.2 Overview of the Developed System

The **TOPBuy App** is a web-based and mobile-responsive data analytics application designed to identify and analyze consumer purchasing behavior through transaction data. The system assists business owners in understanding customer preferences, purchasing frequency, product popularity, seasonal buying trends, and spending habits. By transforming raw sales transactions into meaningful visual reports and predictive insights, the application enables informed business decision-making.

Unlike conventional Point-of-Sale (POS) systems that primarily focus on recording sales transactions, the TOPBuy App integrates data analytics techniques to discover behavioral patterns from historical purchasing data. The application processes customer transaction records and generates dashboards that present statistical summaries, graphical reports, and customer segmentation.

The system aims to provide small and medium-sized enterprises (SMEs) with an affordable business intelligence solution that enhances inventory planning, promotional campaigns, and customer relationship management.

The major objectives of the developed system are as follows:

- Automate the recording and storage of sales transactions.
- Analyze customer purchasing behavior using historical transaction data.
- Generate visual dashboards and analytical reports.
- Identify frequently purchased products.
- Monitor sales trends over time.
- Support business owners in making data-driven decisions.

The development of the TOPBuy App follows modern software engineering principles by incorporating responsive interface design, secure database management, user authentication, and analytical processing.

The completed system consists of several interconnected modules that work together to capture, process, analyze, and visualize business data.

Figure 4.1 illustrates the overall workflow of the developed system.

*(Insert System Workflow Diagram Here)*

The workflow begins with user authentication. Once authenticated, authorized users may record transactions, manage product information, monitor customer records, and access analytical reports generated from accumulated transaction data. The resulting information is presented using interactive charts and statistical summaries to facilitate interpretation by business owners.

### **4.3 System Development Methodology**

The development of the TOPBuy App adopted the **Agile Software Development Methodology**, an iterative approach that emphasizes continuous collaboration, incremental development, and regular evaluation throughout the software development life cycle. Agile was selected because it accommodates evolving user requirements and promotes frequent feedback from stakeholders, ensuring that the resulting application aligns with the operational needs of the intended users.

The Agile methodology enabled the development team to divide the project into manageable iterations, commonly referred to as sprints. At the end of each sprint, a functional component of the system was produced, tested, and reviewed before proceeding to the next iteration. This process minimized development risks and facilitated early detection of software defects.

The development process consisted of the following phases:

#### **Requirements Gathering**

The first phase involved collecting information from prospective users through interviews, observations, and document analysis. Business owners were consulted to determine the common challenges encountered in monitoring customer purchasing behavior and managing sales information. The gathered requirements served as the basis for defining the functional and non-functional requirements of the system.

## **System Planning**

After identifying the project requirements, the researchers established the project scope, objectives, system specifications, development schedule, and resource allocation. The planning stage also included feasibility analysis to determine the technical, operational, and economic viability of the proposed application.

## **System Design**

During this phase, the overall architecture of the application was designed. Database structures, process flows, user interface layouts, and system modules were prepared. Unified Modeling Language (UML) diagrams, Entity Relationship Diagrams (ERDs), and wireframes were developed to provide a blueprint for software construction.

## **Software Development**

The coding phase involved implementing the designed modules using the selected programming languages and frameworks. Individual system functionalities were developed incrementally, including user authentication, product management, transaction recording, customer profiling, analytics generation, and report visualization.

## **Testing**

Each completed module underwent unit testing before being integrated into the entire system. Integration testing ensured that the interaction between modules functioned correctly, while system testing verified the overall performance of the application. User Acceptance Testing (UAT) was later conducted with selected respondents to validate whether the application met user expectations.

## **Deployment**

Following successful testing, the application was deployed in the target environment. Database configuration, server setup, and application installation were performed to make the system accessible to authorized users.

*(Insert the narratives of your deployment here..coordinate with your UI/x)*

## **Maintenance**

The final phase involves monitoring system performance, correcting identified issues, implementing software updates, and enhancing system features based on user feedback. Maintenance ensures the long-term reliability and usability of the developed application.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

*(Insert Agile Methodology Diagram Here or any SLDC model)*

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## **4.4 System Architecture**

The TOPBuy App utilizes a **Three-Tier Client-Server Architecture**, which separates the presentation layer, application layer, and data layer. This architectural approach improves system maintainability, scalability, and security by organizing software components according to their respective responsibilities.

Figure 4.3 illustrates the architecture of the developed system.

*(Insert System Architecture Diagram Here)*

The architecture consists of the following components:

### **Presentation Layer**

The presentation layer serves as the interface between the users and the application. It is responsible for displaying information, receiving user input, and presenting analytical reports through dashboards and interactive visualizations.

Users may access the application using desktop computers, laptops, tablets, or smartphones through a modern web browser.

Major interface components include:

- Login Page
- Dashboard

- Product Management
- Customer Management
- Sales Transactions
- Analytics Dashboard
- Reports

## **Application Layer**

The application layer contains the business logic responsible for processing user requests, validating input data, performing analytical computations, and generating reports.

Its primary responsibilities include:

- User Authentication
- Transaction Processing
- Customer Profiling
- Sales Analysis
- Product Trend Analysis
- Dashboard Generation
- Report Generation

The application server acts as the intermediary between the presentation layer and the database server.

## **Data Layer**

The data layer stores all business information in a centralized relational database. It is responsible for maintaining data integrity, enforcing relationships, and supporting efficient retrieval of information required by the analytics engine.

Stored information includes:

- User Accounts
- Products
- Categories
- Customers
- Sales Transactions
- Transaction Details
- Behavioral Analytics
- Reports

The architecture ensures secure communication between the application server and the database while restricting unauthorized access to sensitive business information.

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**4.5 Hardware and Software Requirements**

The successful implementation of the TOPBuy App requires appropriate hardware and software resources. These specifications ensure that the application performs efficiently during data processing, report generation, and analytics computation.

**4.5.1 Hardware Requirements**

Table 4.1 presents the recommended hardware specifications.

| Component           | Minimum Specification | Recommended Specification |
|---------------------|-----------------------|---------------------------|
| Processor           | Intel Core i3         | Intel Core i5 or higher   |
| Memory (RAM)        | 4 GB                  | 8 GB or higher            |
| Storage             | 256 GB SSD            | 512 GB SSD                |
| Monitor             | 1366 × 768 Resolution | Full HD Display           |
| Internet Connection | 10 Mbps               | 50 Mbps                   |
| Mobile Device       | Android 10            | Android 13 or higher      |

The recommended hardware configuration provides sufficient processing capability for executing analytical operations and rendering graphical dashboards without significant delays.

**4.5.2 Software Requirements**

Table 4.2 presents the software requirements.

| Software                  | Purpose                    |
|---------------------------|----------------------------|
| Windows 11 / Ubuntu Linux | Operating System           |
| Visual Studio Code        | Source Code Editor         |
| MySQL                     | Database Management System |
| PHP / Node.js             | Backend Development        |
| ReactJS                   | Frontend Development       |
| Google Chrome             | Web Browser                |
| Git                       | Version Control            |
| XAMPP                     | Local Development Server   |

The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

#### 4.6 Database Design

The database serves as the foundation of the TOPBuy App by organizing and storing transaction records, customer information, product inventories, and analytical results. A well-designed database ensures efficient data retrieval, minimizes redundancy, and maintains data integrity throughout the operation of the system.

The researchers employed a **relational database model** using MySQL. Database normalization techniques were applied to reduce duplication and maintain consistency among related tables.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.

*(Insert ER Diagram Here)*

The primary entities of the database include:

- Users

- Customers
- Products
- Categories
- Transactions
- Transaction Details
- Analytics Results

## Users Table

This table stores account credentials and user information.

| Field    | Description        |
|----------|--------------------|
| user_id  | Primary Key        |
| fullname | Complete Name      |
| username | Login Username     |
| password | Encrypted Password |
| role     | User Access Level  |

## Customers Table

This table contains customer demographic information used for behavioral analysis.

| Field       | Description     |
|-------------|-----------------|
| customer_id | Primary Key     |
| fullname    | Customer Name   |
| age         | Customer Age    |
| gender      | Customer Gender |



|         |                     |
|---------|---------------------|
| address | Customer<br>Address |
|---------|---------------------|

## Products Table

This table stores product information.

| Field        | Description            |
|--------------|------------------------|
| product_id   | Primary Key            |
| product_name | Product Name           |
| category_id  | Product<br>Category    |
| price        | Selling Price          |
| stocks       | Available<br>Inventory |

## Transactions Table

This table records sales transactions.

| Field            | Description       |
|------------------|-------------------|
| transaction_id   | Primary Key       |
| customer_id      | Foreign Key       |
| transaction_date | Date<br>Purchased |
| total_amount     | Total<br>Purchase |

## Transaction Details Table

This table records the individual items included in each transaction.

| Field | Description |
|-------|-------------|
|-------|-------------|

|                |                    |
|----------------|--------------------|
| detail_id      | Primary Key        |
| transaction_id | Foreign Key        |
| product_id     | Foreign Key        |
| quantity       | Quantity Purchased |
| subtotal       | Item Total         |

## Analytics Table

The analytics table stores computed insights generated by the application. Depending on the implementation, this table may contain aggregated metrics or references to dynamically computed results.

| Field              | Description                        |
|--------------------|------------------------------------|
| analytics_id       | Primary Key                        |
| customer_id        | Foreign Key                        |
| preferred_category | Most Frequently Purchased Category |
| purchase_frequency | Number of Purchases                |
| average_spending   | Average Spending                   |
| generated_date     | Date of Analysis                   |

The relationships among these tables allow the application to trace customer purchases, identify recurring buying behaviors, calculate product demand, and generate comprehensive analytical reports. Proper indexing and foreign key constraints help maintain referential integrity and optimize query performance, ensuring that the system can efficiently support both operational transactions and data analytics.

## **CHAPTER 5**

*(5 to 10 pages..)*

### **CONCLUSION AND RECOMMENDATIONS**

#### **5.1 Introduction**

This chapter provides a comprehensive conclusion to the development of the TOPBuy App. It synthesizes the project's results, assesses whether the research objectives were achieved, and offers recommendations for future enhancements and implementation strategies to further maximize the system's utility for small and medium-sized enterprises (SMEs).

#### **5.2 Summary of Findings**

The development and evaluation of the TOPBuy App yielded significant results aligned with the research objectives:

- The system successfully automated the recording and storage of sales transactions, eliminating manual data entry errors.
- Analytical modules effectively processed historical data to identify purchasing patterns, allowing business owners to predict inventory needs with higher accuracy.
- Visual dashboards provided clear, actionable insights into product popularity and customer segments, directly supporting data-driven decision-making.
- User acceptance testing confirmed that the interface is intuitive, reducing the learning curve for business owners and staff.

#### **5.3 Conclusion**

The TOPBuy App has proven to be a viable and effective solution for SMEs seeking to optimize their business intelligence. By transforming

raw transaction data into behavioral insights, the system empowers owners to make informed strategic decisions regarding inventory management, promotional activities, and customer relationship management. The project successfully addressed the need for an affordable, responsive, and data-centric tool, demonstrating that sophisticated analytics can be made accessible to small-scale enterprises. The research questions were answered affirmatively, confirming that the system enhances operational efficiency and competitive advantage.

## **5.4 Recommendations**

To ensure the continued relevance and efficacy of the TOPBuy App, the following recommendations are proposed:

### **System Improvements**

- Integrate automated notification alerts for low-stock items and expiring products to further improve inventory management.
- Enhance security protocols by implementing two-factor authentication to better protect sensitive sales and customer data.
- Develop advanced export features allowing reports to be generated in multiple formats (e.g., PDF, CSV, Excel) for easier sharing and integration with accounting software.

### **Future Research Directions**

- Explore the integration of machine learning algorithms to enable predictive analytics, such as forecasting future sales trends based on seasonal data.
- Conduct a longitudinal study to analyze the long-term impact of the system on business growth and profitability for SMEs.
- Investigate the feasibility of a mobile-native application to provide offline functionality for businesses with intermittent internet connectivity.

### **Implementation Strategies**

- Business owners should conduct training sessions for staff to ensure consistent data entry habits, which are critical for accurate analytics.
- SMEs are encouraged to perform regular data backups and system maintenance to ensure security and prevent data loss.

## **5.5 Impact of the Study**

The TOPBuy App contributes to the IT field by demonstrating the practical application of data analytics in a resource-constrained environment. By providing SMEs with accessible business intelligence, this study bridges the gap between complex data processing technologies and everyday business operations. While the project faced limitations—such as the need for comprehensive historical data to generate accurate behavioral patterns—the results underscore the significant potential for technology to level the playing field for small businesses in an increasingly data-driven market.

## **5.6 Summary**

In summary, the TOPBuy App represents a successful integration of software engineering principles and data analytics. It serves as a testament to the power of tailored technology solutions in addressing real-world business challenges. This research provides a solid foundation for future developers to build upon, contributing to the broader goal of digital transformation among small and medium-sized enterprises.